ML - system Design I

Lifecycle of ML

1. Model building - Data, model
2. Model evaluation & Experimentation - Precision, Recall, ROC
3. Productionize - Classes
4. Test - Pytest
5. Deployment - AWS (ECR, ECS, Fargate, CI/CD
6. Monitoring & observability

Challenges

1. Browser version changed
2. Data change
3. Change in pattern / trend
4. Data Drift - outdated info → stock market
5. Systems - Req / sec
   - latency
   - CPU / memory / data utilization

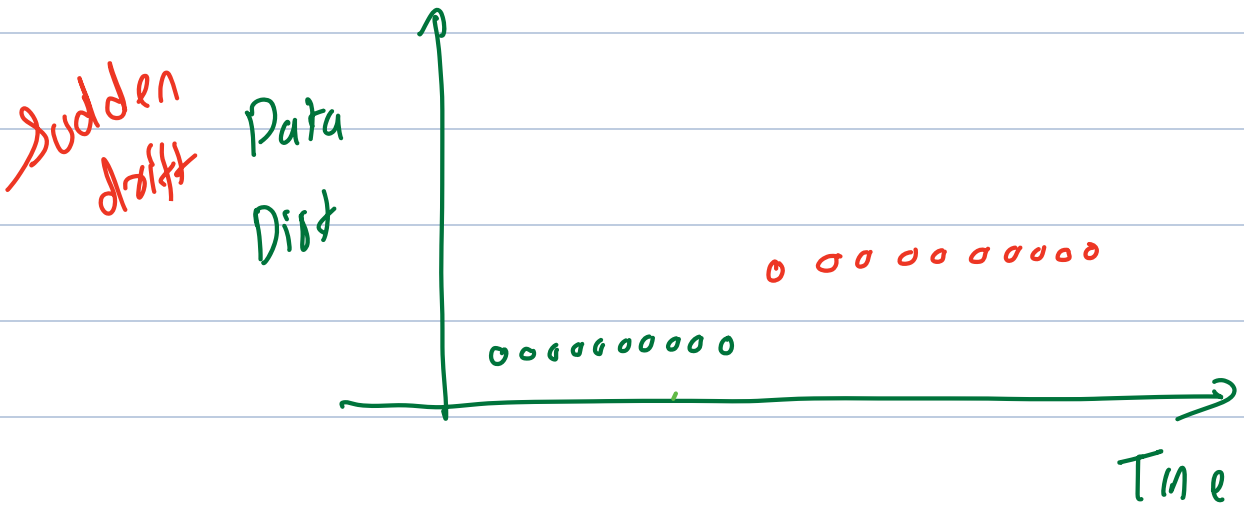
Drift

→ outdated data

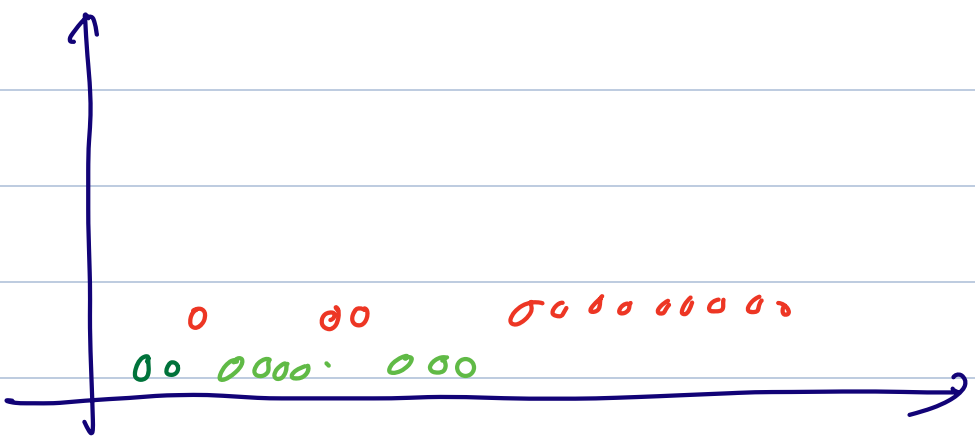
→ Inefficient Model

Type of Drift

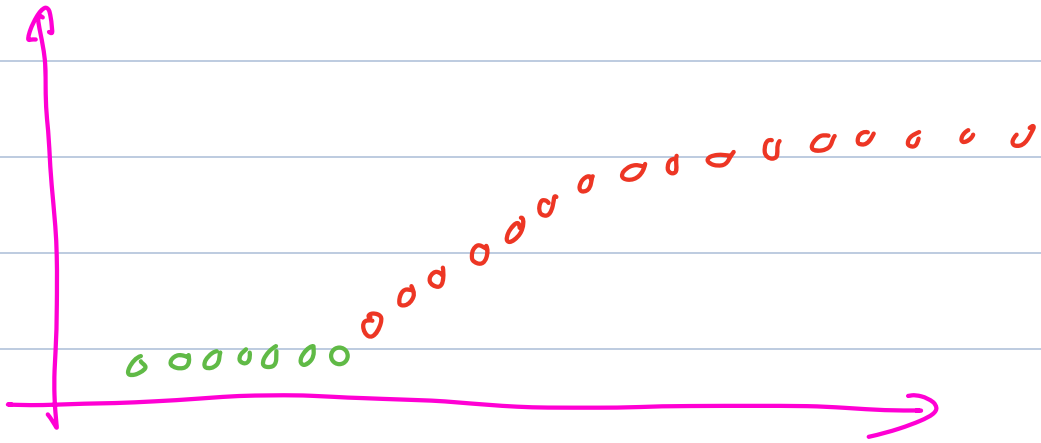
① Concept Drift — model drift



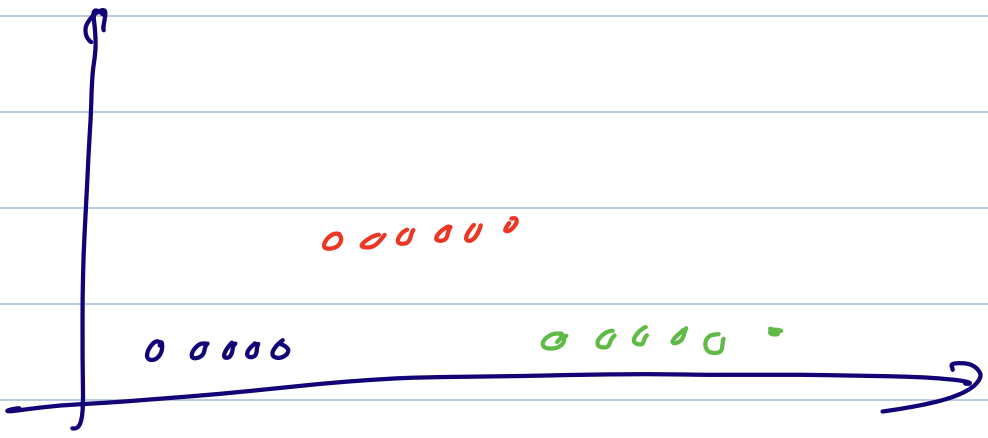
② Gradual Drift



③ Incremental Drift



④ Recurring Drift



② Data Drift

Reasons:

- change in data collection
- —— data source

- ——— in business need
- ——— in distribution

- visual check
- stats test - KS test & chi-square test
- Model performance monitoring
- Data quality check